Data-Driven Modelling Report Lab 1

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I. Introduction

This report explains the work completed for Lab 4 of the DDM course. The main goal was to find a model that is accurate but also simple, meaning it uses as few parameters as possible, for a linear time-invariant system. Using generated input-output data, the system was identified with the Least Squares (LS) method. The best model order was then chosen using three methods: the LS error cost, the Akaike Information Criterion (AIC), and the error on a separate validation dataset.

II. Methodology

The true system is a discrete-time Chebyshev type-II band-pass filter (order 2, passband $\omega \in [0.3, 0.6]$ rad/sample, 3 dB attenuation), truncated to $n_H = 30$ samples forming the exact FIR model $G_0(z)$. Estimation and validation datasets ($N_e = N_v = 300$) are generated as zero-mean, unit-variance Gaussian sequences, with additive noise set to 6 dB or 26 dB SNR. For each order $n = 1, \ldots, 100$, the regressor matrix $\mathbf{H}_n$ is built via `toeplitz` and $\hat{\theta}_n = \mathbf{H}_n \backslash Y_e$. The three model selection criteria are:

$$V_{\text{LS}} = \frac{1}{N_e \sigma_v^2} \|Y_e - H_n \hat{\theta}_n\|^2 \tag{1}$$

$$V_{\text{AIC}} = V_{\text{LS}} \left(1 + \frac{2n}{N_e}\right) \tag{2}$$

$$V_{\text{val}} = \frac{1}{N_v \sigma_v^2} \|Y_v - H_n(u_{0v}) \hat{\theta}_n\|^2 \tag{3}$$

Robustness is assessed over 100 Monte Carlo runs with fresh noise realizations at each SNR level.

III. Results and Discussion

A. *Model Order Selection at 6 dB SNR*

Figure 1 shows the model selection criteria for the normal SNR case (6 dB). The training cost V_{LS} decreases monotonically with increasing model order, reaching its minimum at $n = 100$. This behavior is expected since adding more parameters always improves the fit to the training data, even when fitting noise. In contrast, both V_{AIC} and V_{val} exhibit clear minima at approximately $n = 16$. The AIC penalty term $1 + 2n/N_e$ effectively counteracts the decreasing trend of V_{LS}, while the validation cost naturally penalizes overfitting through independent data. This demonstrates that the effective model order of the system is lower than the truncated length $n_H = 30$, as the later impulse response coefficients contribute negligibly to the output given the noise level.

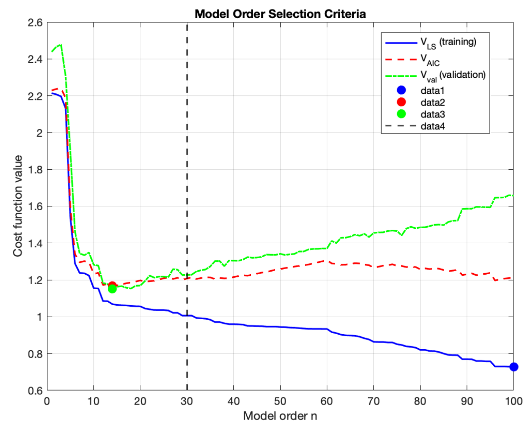

Fig. 1: Model selection criteria (6 dB SNR)

B. *Effect of Reduced Bandwidth*

Figure 2 shows the model selection criteria after reducing the bandwidth of the Chebyshev filter while keeping the SNR at 6 dB. The training cost V_{LS} continues to decrease monotonically, reaching its minimum at $n = 100$, confirming that least squares alone remains unsuitable for model order selection regardless of the system bandwidth. In contrast, both V_{AIC} and V_{val} now exhibit their minima at approximately $n \approx 30$, which coincides with the truncated impulse response length n_H. This is a significant shift compared to the original bandwidth case, where both criteria selected $n \approx 16$.

This behavior can be explained by the relationship between bandwidth and impulse response duration. A narrower bandwidth in the frequency domain corresponds, by the time-frequency duality, to a slower decay and thus a longer impulse response in the time domain. As a result, more FIR coefficients are required to capture the system dynamics accurately, and the effective model order increases toward n_H. This confirms that the optimal model order is not solely determined by the noise level, but also by the spectral complexity of the underlying system.

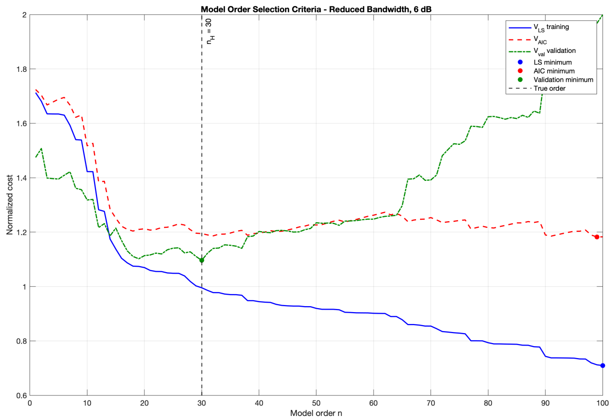

Fig. 2: Model selection criteria (Reduce Bandwidth)

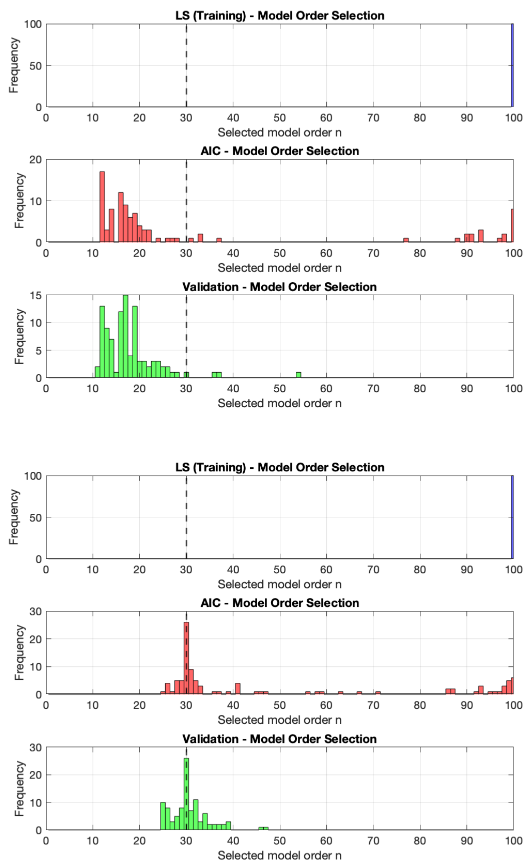

Fig. 3: Robustness histograms at 6 dB (top) and 26 dB (bottom).

C. Robustness Analysis

Figure 3 show the histograms of selected model orders over 100 Monte Carlo runs at 6 dB and 26 dB SNR respectively. In both cases, V_{LS} deterministically selects $n = 100$ for every run. V_{AIC} and V_{val} show narrow distributions centered around $n = 16$ at 6 dB and $n = 30$ at 26 dB, consistent with the single-run results. The wider spread at 26 dB reflects the larger number of identifiable parameters, yet both criteria remain reliable across all noise realizations.

D. Effect of Increased SNR (26 dB)

Figure 4 displays the model selection criteria for the high SNR case (26 dB). The V_{LS} curve again decreases monotonically to $n = 100$. However, the minima of V_{AIC} and V_{val} have shifted to approximately $n = 30$, which coincides with the true truncated impulse response length n_H. With less noise, the weaker coefficients in the tail of the impulse response become identifiable, allowing AIC and validation to select a higher model order that better captures the true system dynamics. This confirms that SNR directly influences the optimal model order: higher SNR enables estimation of more coefficients.

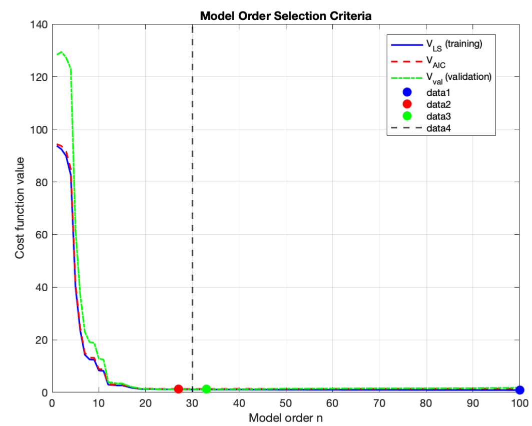

Fig. 4: Model selection criteria (26 dB SNR)

IV. Conclusion

The results clearly demonstrate that V_{LS} alone is unsuitable for model order selection, as it always selects the maximum tested order. Both AIC and validation successfully identify parsimonious models, with the selected order increasing as SNR improves. The robustness analysis confirms that these methods yield consistent selections across multiple noise realizations, making them reliable tools for practical system identification.